

Auditing Frozen Foundation-Model Embeddings: A Leakage-Controlled, Calibration-Aware Evaluation Protocol Stress-Tested on Single-Cell Genomics

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Abstract—Evaluating frozen foundation-model (FM) embeddings is unreliable when the scoring metrics are entangled with the methods they rank, when tokenizer coverage is confounded with representation quality, and when the exchangeability that underwrites uncertainty guarantees is broken by distribution shift. We present a reusable, deterministic evaluation protocol for pretrained representations under distribution shift, built on four domain-general moves: every agreement metric is paired with a non-linear (k NN) probe and reference-free structure metric to expose label circularity; tokenizer coverage is treated as a measured variable; parity is argued with an equivalence test (TOST) rather than an absent p -value; and reliability is audited with split-conformal coverage, ECE calibration, and selective abstention. The protocol’s signature move replaces binary leaderboard verdicts with a dose–response analysis that places the criticized method class on the curve, converting a rank into a manipulable, causal mechanism. We stress-test it on single-cell genomics—a regime of batch/donor shift, circular annotation, and cross-species tokenizer mismatch—across four modalities, up to 24 leakage-controlled atlases, and five FM families. Three findings follow: zero-shot single-cell FMs neither beat nor trail simple baselines on matched vocabularies; the much-warned cross-batch calibration collapse is a general exchangeability failure of the data regime, not an FM defect; and which side “wins” is set by task and metric, not model scale or architecture. All findings are scoped to the zero-shot, frozen-embedding regime on coarse human cell types; fine-tuning and few-shot transfer are untested and can reverse them.

Index Terms—representation evaluation, frozen foundation-model embeddings, distribution shift, evaluation-metric circularity, conformal prediction, calibration, equivalence testing, dose–response analysis, single-cell genomics.

I. INTRODUCTION

Foundation models [1], [2] are deployed as *frozen* feature extractors: pretrained on unlabelled corpora, they embed new inputs without task-specific training and are ranked on leaderboards against cheaper baselines. Evaluating them fairly is unsettled—leaderboards use agreement metrics that presuppose the ground truth, rarely auditing confounding or reliability under shift. Single-cell genomics is a sharp stress test: models pretrained on millions of cells [3], [4] (scGPT [5], Geneformer [6], scFoundation [7], UCE [8], CellPLM [9], spatial/chromatin variants [10], [11], [12]) are promised to recover cell types, integrate batches, and support inference better than linear pipelines. A growing critical literature disputes this in

the zero-shot setting, one paper and one task at a time (Sec. II), leaving the reliability axis—calibration, conformal coverage, selective abstention—almost entirely unaudited; no calibration or conformal audit of any scATAC foundation model exists.

These disputes share a flaw not specific to genomics: the yardsticks are entangled with the methods they rank. Three confounds recur. (i) *Label circularity*: when a benchmark’s “ground truth” is derived in the representation space it grades, agreement scores reward proximity to the label-generating process, not the signal—cell-type labels come from clustering and marker scoring [13] in the same expression space a linear baseline like PCA occupies, so ARI/NMI/scIB/label-AUROC can reward recovering the clustering that defined the labels. (ii) *Tokenizer coverage*: a fixed vocabulary silently drops unmappable inputs (gene symbols vs. Ensembl IDs, cross-species orthologs), scoring *coverage* while appearing to score representation. (iii) *Broken exchangeability*: split-conformal sets are valid only under exchangeability [14], which batch and donor shift routinely violate. Genomics makes all three severe at once.

We present the evaluation protocol as the contribution, genomics as the stress-test domain: a deterministic, leakage-controlled fair-evaluation and reliability audit for frozen embeddings under shift. It pairs every agreement metric with a non-linear (k NN) probe and reference-free structure metric, treats vocabulary coverage as a measured variable, argues parity with an equivalence test (TOST), and audits reliability (split-conformal coverage, ECE calibration, selective abstention)—its signature move replacing a binary leaderboard verdict with a graded *dose–response* that places the criticized method class on the curve, turning a rank into a causal mechanism.

Applied to genomics, the protocol yields three provisional findings: on vocabulary-matched atlases frozen embeddings show no advantage *or* deficit over simple baselines (the apparent “win” dissolving into label circularity and a causal tokenizer-coverage threshold); the cross-batch calibration collapse is a *general* exchangeability failure of the data regime, not an FM defect (and is absent in scATAC); and which method “wins” is set by task and metric, not model scale or architecture. We also report the first single-cell ATAC calibration/reliability audit and the first zero-shot spatial-FM head-to-head against a parameter-free niche baseline (Fig. 1).

a) *Contributions.*: The deliverable is an evaluation *instrument*, not a verdict on foundation models: (i) a fair, leakage-controlled, reliability-aware protocol for frozen em-

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beddings under shift, scoring a representation axis (non-linear probe, reference-free structure metric, equivalence testing) and a reliability axis (conformal coverage, calibration, abstention) on identical splits; (ii) dose–response in place of binary verdicts, placing the criticized method class on the curve; (iii) three deconfounding tools that generalize beyond genomics (non-linear probe with reference-free metric, coverage-as-measured-variable, equivalence testing); and (iv) a genomics stress test—four modalities, up to 24 leakage-controlled atlases, five FM families, a 12-study meta-analysis—with two modality firsts (the first scATAC calibration/reliability audit and the first zero-shot spatial-FM vs. parameter-free niche baseline). **Scope:** all findings hold in the zero-shot, frozen-embedding regime on coarse human cell types; fine-tuning and few-shot transfer are untested and can reverse them.

II. RELATED WORK

Single-cell foundation models pretrain a transformer on millions of unlabelled cells, exposing a frozen feature extractor for annotation, integration, and perturbation inference [2], [1]. The first wave spanned modalities—transcriptome encoders [5], [6], [7], a cross-species protein-embedding tokenizer [8], a generative latent model [9], and spatial/chromatin variants [10], [11], [12], [15]—all claiming pretraining recovers biology better than the linear pipelines they replace. We take these frozen encoders as the object of measurement.

A second camp reports the opposite in the zero-shot setting, and our protocol is most easily confused with it: each is a careful but *single-task* verdict on a *single* yardstick—zero-shot embeddings do not beat PCA on clustering agreement [16], a linear probe rivals a far larger model [17], “one PCA still rules them all” on perturbation [18], and figure-level [19] or biology-driven [20] readings reach the same skeptical conclusion.

The perturbation literature converges independently: simple additive/linear controls match or exceed deep and foundation predictors [21], [22], [23], [24], reproduced at benchmark scale [25], [26], [27] and against graph baselines [28]; attention captures co-expression rather than regulatory signal [29], and pretraining scale/diversity add little [30]. We read this as vote-counted context [31], not a pooled effect—all sharing one *method*: a binary rank on one agreement score, untested for confounding and unaudited for deployment.

The confound sits in the yardstick itself: integration/annotation are scored by label-agreement suites—scIB/ARI/NMI panels [32], consensus/hierarchical annotation pipelines [33], [34], deep-integration benchmarks [35]—whose “ground truth” comes from clustering and marker scoring [13], [36] in the same expression space a linear baseline occupies. Agreeing with such labels rewards recovering the label-generating partition, not the signal—a circularity overcorrection-aware batch metrics have begun to flag [37]. No critique pairs its score with a label-independent structure check, treats tokenizer coverage as measured, or argues parity by equivalence, so a coverage or coarse-label artifact is indistinguishable from a representational deficit.

The reliability axis is almost absent from both camps: split-conformal prediction gives distribution-free coverage [38],

[39], but only under exchangeability, which single-cell batch/donor shift routinely breaks [14], and neural confidences are miscalibrated without correction [40]. Single-cell uptake is recent and baseline-only: conformal annotation [41] and hierarchical reject options [42] audit classical classifiers, not frozen FM embeddings. The gap widens outside transcriptomics—spatial and chromatin models are evaluated almost only on in-modality label recovery, and to our knowledge no calibration/conformal audit of a scATAC FM, nor a zero-shot spatial FM against a parameter-free niche baseline, has been reported. A leaderboard rank says nothing about whether uncertainty is trustworthy under shift—the property deciding whether an embedding is safe to deploy.

Our contribution folds these observations into one reusable instrument rather than adding another verdict; the framing is domain-general, with single-cell genomics the stress test that makes all three confounds severe at once.

III. DATA AND METHODS

A. Corpus

We catalogued 37 entries (~30 benchmark studies plus infrastructure and counter-evidence); 12 gave usable per-comparison numbers (10,759 comparisons; machine-readable tables where released, else source-tagged figure/supplementary values). One non-peer-reviewed/AI-authored item was dropped (criterion in the extractability manifest).

B. Pooling

Within each (study, task, dataset, metric) group g : every simple-baseline b paired against every FM/DL competitor m , scored via win indicator $w = \mathbb{I}[s_b \succ s_m]$ under the authoritative per-metric direction $\succ$ (ties = 0.5), giving per-study win-rate $\hat{p}_s = \frac{1}{|C_s|} \sum_{c \in C_s} w_c$ over study s ’s comparisons C_s . Since scPerturBench [26] alone supplies ~96% of raw comparisons (>99% in the perturbation cluster), the headline estimator: equal-weight-per-study mean $\bar{p} = \frac{1}{S} \sum_{s=1}^S \hat{p}_s$, 95% CI via nonparametric bootstrap over the S studies (10,000 replicates, seed fixed); naive per-comparison rate kept for contrast.

Rows are tagged by method family (mean/linear/classical-DR vs. mlp/FM/DL) and metric family (correlation/MSE/distribution; DEG-only vs. all-genes) for stratification. This is a deliberately conservative *vote-counting* synthesis, scoring direction rather than a common effect size—which the heterogeneous metrics (ARI, MSE, AUROC, correlation) do not share—at a cost in magnitude, per-study precision, and power vs. an effect-size meta-analysis [31]. We treat it as descriptive context only, never sole basis for a load-bearing claim (those rest on the self-computed re-derivations of clusters J–L, Table I).

C. Fair-evaluation core and dose–response designs

To deconfound the label-matching critique we score every representation three ways: (i) a *non-linear* probe—distance-weighted k NN ($k=15$) on the held-out largest batch—avoiding the bias of the naive *linear* comparator (multinomial logistic regression, cross-batch macro-AUROC), which flatters PCA’s

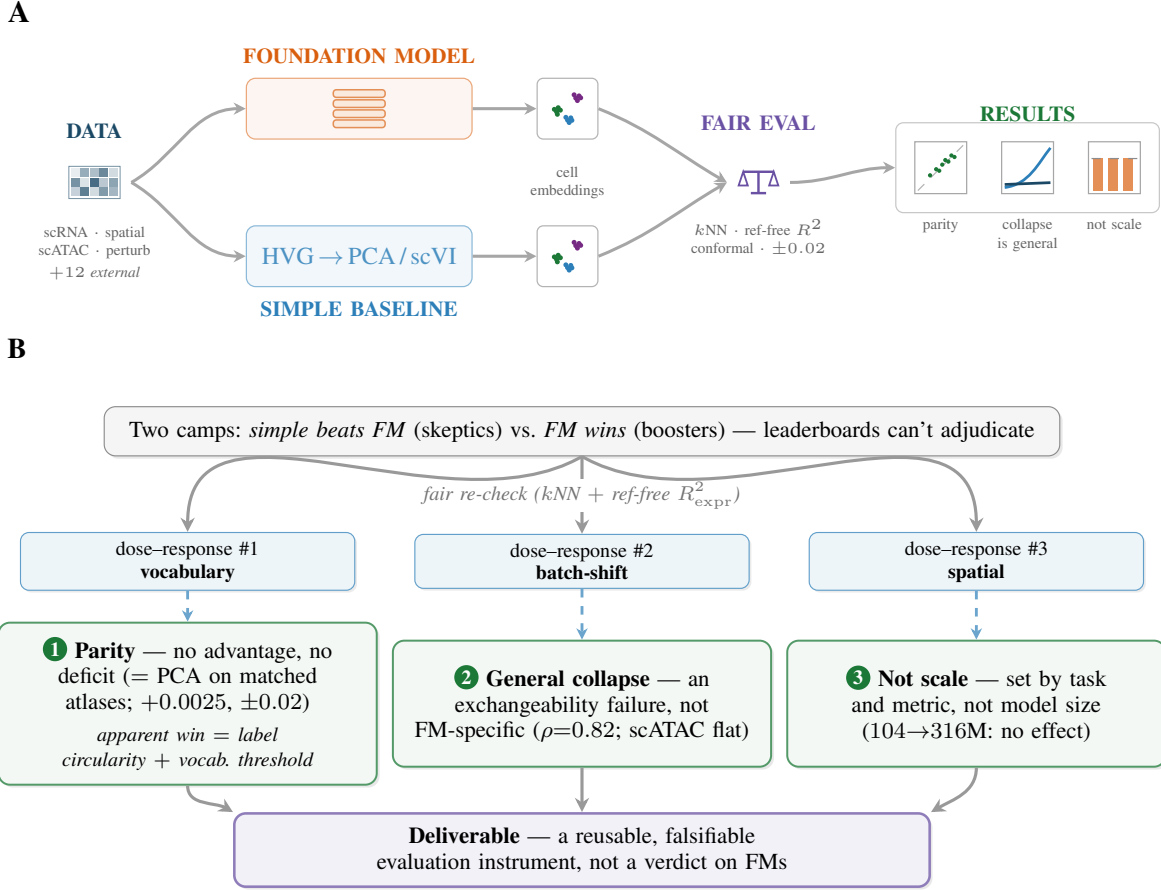


Fig. 1. *Paper at a glance.* (A) Study architecture: four self-computed modalities (scRNA, spatial, scATAC, perturbation) plus an external 12-study meta-analysis are embedded by foundation models *and* simple baselines, then scored by a fair-evaluation suite (non-linear probe, reference-free structure metrics, split-conformal coverage, an equivalence test, three dose-response analyses). (B) Argument roadmap: a reference-free re-check turns the two-camp dispute into three findings—the apparent baseline “win” dissolves into two artifacts (label circularity, a causal vocabulary threshold), with three dose-response probes (vocabulary coverage, batch-shift strength, spatial-aggregation depth) supplying the mechanisms.

shared linear space with the labels; (ii) a *reference-free* structure metric, R^2_{expr} (spatial niches: R^2_{compo})—the fraction of highly-variable-gene expression (or neighbourhood cell-type-composition) variance explained by a method’s *own* unsupervised K -means partition ($K = \text{ground-truth label count}$), no labels used, scoring structure not label agreement; where instead scored *against* labels (e.g. the spatial-niche ARI below) we use the adjusted Rand index (1 = identical, 0 = chance).

(iii) An *equivalence* test for parity: per FM family, bootstrap the mean paired per-atlas difference from PCA against a pre-stated ± 0.02 margin (TOST-style)—“no advantage” as a positive equivalence result, not an underpowered failure-to-reject—checked by a two-sided Wilcoxon signed-rank test on paired atlas-level differences and, where arms share an atlas set, by an atlas-cluster bootstrap (resampling atlases, the unit of independence, not arm $\times$ atlas rows). We never use a best-of- N statistic (max-selection bias).

Each mechanism is traced as a *dose-response*: apparent FM quality vs. tokenizer-coverage fraction (an atlas’s gene-symbol/ID share of a given FM’s vocabulary—the objective, performance-blind criterion splitting atlases vocabulary-matched vs. -mismatched, near-zero for cross-species/non-symbol IDs, else substantial), probed by a causal within-atlas gene ablation fixing the atlas and removing only read-

able genes; coverage collapse vs. batch-shift strength (cross-validated AUROC, held-out-batch membership) across two degradation metrics, FMs *on* the curve; niche prediction vs. spatial-smoothing dose. Modality/regime contrasts (e.g. scRNA-vs-scATAC coverage-gap slope) use an interaction term, linear regression of gap on shift strength (slope-difference t -test), corroborated by a label-permutation test; spatial probes use spatially-blocked cross-validation (avoiding leakage).

D. Reliability machinery

Calibration uses Expected Calibration Error, $ECE = \sum_{j=1}^B \frac{|B_j|}{n} |\text{acc}(B_j) - \text{conf}(B_j)|$, over $B=15$ equal-width bins. Readout capacity: three frozen-embedding classifiers—multinomial logistic regression (linear), MLP-64, MLP-256—trained on the calibration split, scored cross-batch on the held-out batch, isolating decoder capacity from the representation. Temperature scaling [40]: fits scalar T minimizing negative log-likelihood on a held-out split, rescaling logits (z/T) before test-time softmax.

Predictive uncertainty: split-conformal LAC sets—calibration scores $\alpha_i = 1 - \hat{\pi}_{y_i}(x_i)$, threshold $\hat{q} = \lceil (n_{\text{cal}}+1)(1-\alpha) \rceil / n_{\text{cal}}$ quantile, test set $C(x) = \{y : 1 - \hat{\pi}_y(x) \leq \hat{q}\}$ —giving marginal coverage

$\geq 1 - \alpha = 0.90$ [39]. Coverage gap under shift: in-distribution minus shifted coverage (positive = under-coverage); baseline: the random/matched split (nominal 0.90) for cross-sample shift, control-to-control for cross-diagnosis (Control $\rightarrow$ AD) shift.

The confidence $\leftrightarrow$ correctness link: a partial correlation deconfounded by log total fragments, log peak count, and sample via Frisch–Waugh–Lovell (regress both on confounders, correlate residuals). Selective performance: AURC over the risk-coverage curve; negative controls shuffle labels and permute features (both must collapse to chance); all CIs are 1,000 $\times$ bootstrap.

E. Self-computed audits

Spatial niche identification: a CosMx human lymph-node dataset [43], three zero-shot spatial FM—scGPT-spatial [44], Novae [45], Nicheformer [10]—run locally via self-written loaders bypassing uninstallable packages, figshare/HF weights loaded exactly. scATAC cell-type calibration: GSE174367 snATAC [46] (130,418 cells, 20 samples, 7 brain types), leakage-controlled cross-sample and cross-disease (Control $\rightarrow$ AD) splits.

ChromFound [11]: run zero-shot from official modules and pretrained weights (loaded exactly); only the CUDA-only Mamba selective-scan and flash-attention kernels swapped for numerically-equivalent reference implementations for GPU compatibility (official architecture, not a re-implementation), on a disclosed top-2048-peak/20k-cell subset. Atacformer [12] (standard transformer encoder, 890k-region hg38 universe): added via a self-written loader mapping GSE174367 peaks to universe tokens by genomic overlap (191,858/219,070 mapped; 0 missing/0 unexpected weights). EpiFoundation [15] remains unrun (env-incompatible kernels), a disclosed limitation.

The self-computed scRNA suite (clusters J–L): local leakage-controlled atlases, frozen FM embeddings, locally-computed classical baselines (PCA/HVG/scVI [47]), leave-one-batch-out splits, same calibration/conformal/abstention machinery. Batch integration scored by LISI (local inverse Simpson’s index, each cell’s k NN graph, fixed perplexity): iLISI on batch identity (higher = better mixing), cLISI on cell-type identity (higher = more mixing, i.e. worse purity).

The perturbation arm: GEARS [28] on Norman-2019 [48] against an additive baseline predicting control $+\delta_A + \delta_B$ from train-only single-perturbation deltas δ_A, δ_B ; double perturbations stratified by combo novelty (seen2/seen1/seen0: both/one/neither single perturbation seen in training), scored by full-gene MSE (all measured genes, no oracle/DE subselection).

The expansion sweep adds k NN, nearest-centroid, and random-forest to the simple-baseline panel across 24 annotated atlases from a 95-atlas local inventory (cell-type + a ≥ 2 -level batch field), plus self-embedded zero-shot FMs: Geneformer-V2-104M (package bypassed; faithful rank-value tokenization, mean-pooled over gene tokens) and, for the fair multi-FM check, four more arms across three families—Geneformer-V2-316M, scFoundation, CellPLM, UCE (vendored code / released eval pipelines).

The FM arm grows from 3 to 20 raw-count atlases, transferring labels from each curated (normalized) atlas onto its raw-count GSE twin via cell-barcode matching; baselines are recomputed on log-CP10k-normalized counts for fairness. Seeds fixed; thread-capped for determinism.

F. Reproducibility

All results derive from deterministic scripts (fixed seeds) over released tables and local data; conformal coverage agrees to 16 digits across a custom implementation, an independent re-derivation, and two public libraries [49], [50]. Self-computed audits passed adversarial verification, correcting an initial scATAC-FM overclaim to the matched-control statement reported here.

The two most-relied-on scRNA FM loaders are validated against official code. scGPT reproduces the official cached embedding to three decimals. Geneformer-V2 reproduces the official tokenizer exactly (cosine 1.000, bone marrow); the official 4096-token budget gives the same k NN-AUROC as our 2048 (0.872 vs. 0.871)—faithful tokenization, a pessimistic shorter budget. Atacformer was checked against official geniml code: a plain transformer encoder (no positional embeddings), reconstructed by our loader; a GeLU-vs-official-ReLU feed-forward difference shifts the embedding (cosine 0.93) but not cross-sample accuracy (0.962 vs. our 0.965, both below LSI). Since the official class does not instantiate on the current transformers release, we used a faithful replica with the official weights.

We assign every self-computed scRNA claim (clusters J–L) a confidence level—supported, qualified, withheld—phrased conservatively (e.g. “residual benchmark-correctness signal,” not “biological signal”): cluster J is reproduction (qualified for the second FM); cluster K’s coverage-collapse is best-supported (wording scoped to benchmark correctness); cluster L’s additive-beats-GEARS result is now supported by a label-fair full-gene comparison (superseding the oracle-gene version), only the selective-prediction half still exploratory. One earlier parser-bug-affected result was flagged and excluded. A provenance manifest—SHA-256 of every paper-backing table and script, plus the captured environment—lets any table be checked bit-for-bit against a re-run.

IV. RESULTS

Results are cross-indexed by *cluster* (A–L), each mapping an external or self-computed arm to its task; the full mapping is in Table I, and the self-computed clusters are H–L (the scRNA suite is J–L).

A. Representation, annotation, integration: no FM advantage

Across five studies [16], [17], [20], [19], [18], simple baselines beat zero-shot scFMs in 88.7% of studies (84.6% of 331 comparisons; 95% CI [0.714, 1.00]); classical dimensionality reduction (HVG/PCA/scVI/Harmony) wins 96.7% [0.901, 1.00] (Fig. 2). One, Atti & Subramaniam [19] (entry A4, tier-C), is scored *qualitatively* as a single 1.00 study; dropping it barely shifts the win-rate (0.887 $\rightarrow$ 0.86; values reused from each study’s public release).

TABLE I

CROSS-CLUSTER SUMMARY. WIN-RATE IS THE EQUAL-WEIGHT PER-STUDY MEAN (95% CI BY STUDY-LEVEL BOOTSTRAP); k = NUMBER OF STUDIES. CLUSTERS ARE KEPT SEPARATE (DIFFERENT TASKS/METRICS; NOT POOLED INTO A SINGLE NUMBER). E AND F ARE ONE MERGED GRN CLUSTER, SO THE TWELVE LABELS A–L ARE ELEVEN CLUSTERS; ROWS FOLLOW THE EXTERNAL-VS-SELF-COMPUTED GROUPING OF THE FOOTNOTE, NOT ALPHABETICAL ORDER.

Cluster	Claim (zero-shot)	Win-rate [95% CI]	Strength
A	no FM advantage (rep./annot./integ.)	0.887 [0.71, 1.00], $k=5$	circular-metric [†]
B	linear/additive $\geq$ DL/FM (perturbation)	0.653 [0.37, 0.87], $k=7$	metric-contingent
H	no spatial-FM advantage (niche-ID), fair test	FM kNN 0.76–0.86	circular-metric [†]
E/F	raw scFM signal $\leq$ gene-level baselines (GRN)	3/3, $k=1$	usage-dependent
G	trajectory/velocity: no scFM evaluated	—	open gap
D	scIB/kBET/LISI unreliable at scale	—	caveat
C	ensemble/calibration give reliable UQ; FM UQ untested	—	supportive
I	scATAC calibration robust to shift; 2 FM, no gain over LSI	self-computed, 2 FM	first audit
J	zero-shot scFM = baseline (parity), fair test	FM ties/beats PCA, 11 matched	reproduction+
K	cross-batch coverage collapse is general (not FM-specific)	20/24 atlases	novel (reframed)
L	additive $>$ GEARS (perturbation, fair full-gene); FM UQ modest	3/3 strata	supported

Clusters A–G aggregate or audit external results; H and I are self-computed audits (spatial, scATAC) and J–L are self-computed scRNA reproductions/audits on local atlases. [†]Clusters A and H rest on label-matching metrics (ARI/scIB, and spatial-niche ARI) with a circularity our fair re-checks expose (via cluster J and cluster H's own composition re-check, respectively); read as “no FM advantage,” not a decisive baseline win.

Almost all these are label-matching metrics (ARI, NMI, scIB) scored against clustering/marker-derived labels in the same expression space as the linear baselines. Our re-analysis (cluster J, Fig. 3) finds two inflating artifacts: label-matching circularity, and gene-vocabulary mismatch—cross-species (mouse-symbol) and gene-ID-format (ENSG) atlases register as FM failures only because the human-symbol tokenizer maps almost none of their genes. Controlling both, baseline and FM are *tied* on vocabulary-matched atlases (Table III).

So cluster A shows no FM advantage or deficit: zero-shot scFMs neither beat nor trail far cheaper baselines on vocabulary-matched coarse cell types, not a decisive baseline win.

B. Perturbation prediction: real but metric-contingent

Pooled equal-weight win-rate is 65.3% [0.372, 0.872], falling to 51.8% [0.151, 0.867] for studies with ≥ 3 comparisons—both wide, both *including parity* (0.50). Cluster B rests on only $k=7$ studies (4 restricted): an underpowered “no clear winner,” not a baseline advantage. Baselines win on delta/ L_2 /distribution metrics (Csendes [23], Kernfeld [22], Wong–Pfizer [21] at 1.00; Ahlmann-Eltze [24] 0.60) but FM/DF compete on absolute-correlation metrics (PertEval [25] 0.00): which side wins depends on the metric family.

C. Spatial niche identification: no FM advantage after a fair re-check

On the CosMx lymph node (19,718 cells, 4 niches) [43], unsupervised ARI makes three zero-shot spatial FM look near-useless (0.386 nbhd-composition baseline vs. scGPT-spatial 0.139, Novae 0.124, Nicheformer 0.012)—but a fair re-check (Fig. 4) overturns most of this. The niches are *composition-defined* (R^2_{compo} 0.234, near-zero single-cell R^2_{expr} 0.007),

so a per-cell FM is scored against a label with no per-cell signal while the nbhd-composition baseline—the niche-definition recipe itself—wins near-tautologically.

A *supervised kNN* niche-probe (spatially-blocked CV) finds all three FM carry niche information (AUROC 0.76–0.86), far above their 0.01–0.14 ARI—the unsupervised score was a clustering/metric artifact. A *residual, real* gap remains: the FM still trail simple spatial baselines (spatial-smoothed PCA 0.92, nbhd-composition 0.90; Fig. 4C peaks 0.93)—no FM advantage, but not the “near-useless” ARI over-stated.

Dosing spatial context localizes the gap. A per-cell model here lacks only spatial aggregation: smoothing a per-cell PCA over its k nearest neighbours raises niche prediction monotonically (Fig. 4C–D), from FM level at $k=0$ (AUROC 0.835, macro-F1 0.40; raw-FM AUROC scGPT-spatial 0.826, Novae 0.862) to AUROC 0.93 / F1 0.76 (Spearman $\rho = 0.98$ for F1)—*overtaking all three FM and the composition baseline* by $k \approx 25$ –50—before over-smoothing erodes it past $k=100$. The same dose lifts scGPT-spatial too (F1 +0.20, $\rho = 0.93$): the “spatial-FM gap” is one missing operation, spatial aggregation, supplied by PCA smoothing, not a representational deficit.

Two checks bound this claim: smoothing helps per-cell PCA most (macro-F1 +0.36) and the strongest raw FM (Novae) least (+0.06, as it already encodes neighbourhood context), and an unsupervised check is *null* (smoothed-PCA R^2_{compo} peaks at $k=6$ then declines), so we anchor the claim to the supervised probe. The demonstration is single-dataset (one CosMx lymph node); a strictly fold-blocked recompute preserves the overtake (smoothed-PCA macro-F1 0.70 vs. raw FM 0.34–0.53), so it is not a cross-fold-leak artifact, though bounded to one dataset.

D. GRN, trajectory, integration metrics, and uncertainty

GRN: raw scFM signal (Geneformer attention AUROC 0.70) loses to gene-level baselines (variance 0.88, mean-

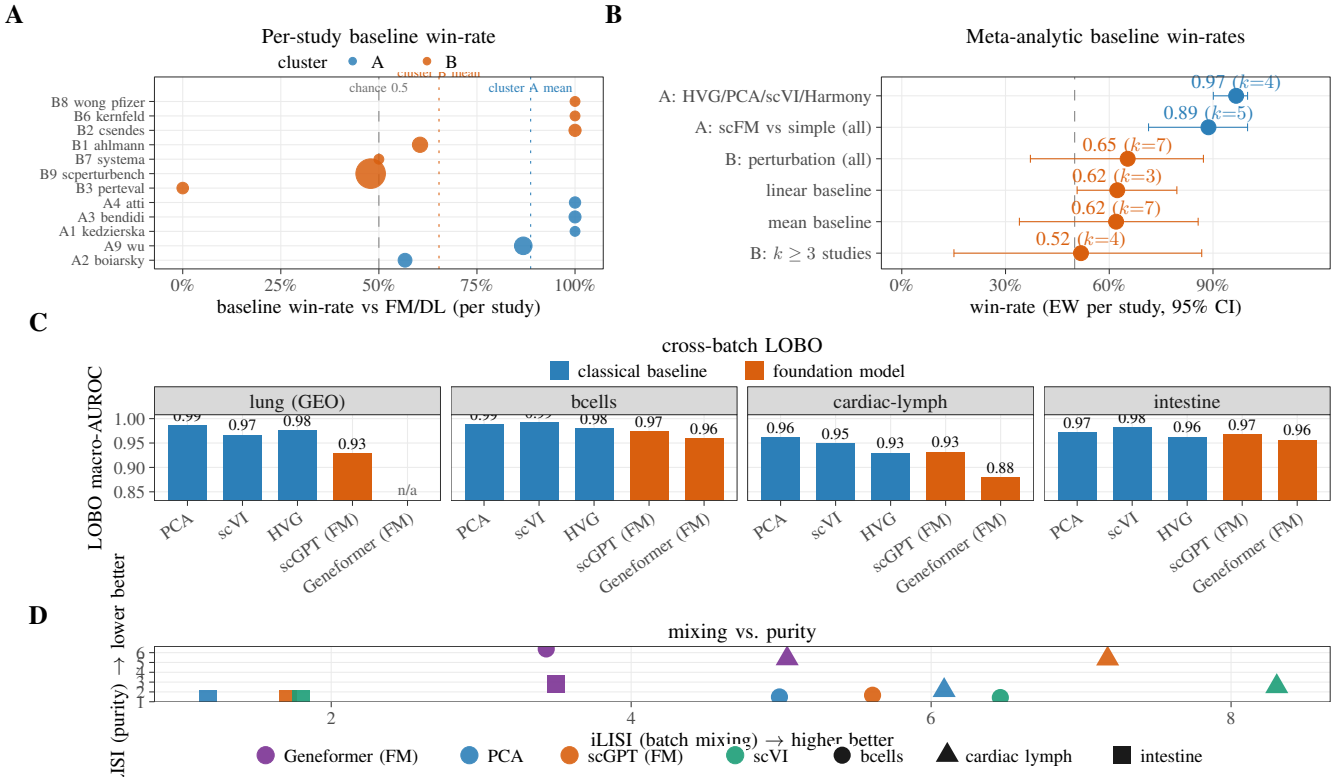


Fig. 2. *External context: meta-analysis, LOBO reproduction, and integration trade-off* (clusters A–B, J). (A) Per-study baseline win-rate vs. FM/DL (point size $\propto$ #comparisons). (B) Meta-analytic baseline win-rates by stratum (equal-weight per study; 95% CI over studies). (C) Self-computed (cluster J): on four leakage-controlled scRNA atlases, frozen zero-shot scFMs do not dominate classical baselines under leave-one-batch-out; PCA or scVI is best in every atlas. (D) Integration trade-off: the one axis FMs “win”—batch mixing (higher iLISI)—costs cell-type purity (higher cLISI); the purpose-built scVI dominates the mixing/purity trade-off (strictly best on both axes for B-cells, Pareto-front for the other two), so even the apparent integration advantage is no all-things-considered win (per-atlas iLISI gain +0.6/+1.1/+2.4 paired with cLISI cost +0.1/+3.1/+1.5; three atlases, an observation not a dose-response).

expression 0.84, 1–dropout 0.81; 3/3) [29]; FM embeddings help only when fed to a *supervised* GNN (scRegNet [51]; usage-dependent). *Trajectory/velocity*: a 15-method $\times$ 17-dataset benchmark [52] contains *zero* foundation models—an open gap. *Integration metrics*: kBET/iLISI power collapses to zero under large batch effects and scIB [32] is gameable [37], [35], caveating all scFM scIB comparisons (ours included). *Uncertainty*: ensemble-consensus and calibration give reliable uncertainty (popV [33] consensus 8 $\rightarrow$ 95% correct; hierarchical-loss annotation with a reject option [34], [42]), but FM-specific uncertainty remains unaudited.

E. scATAC-seq reliability: a first audit and a modality contrast

The FM-free peak-LSI cell-type probe is well calibrated (ECE 0.006), with split-conformal coverage holding under both cross-sample (gap -0.007) and cross-disease Control $\rightarrow$ AD shift (gap -0.028)—unlike scRNA scGPT, where cross-batch coverage collapses by 0.123 (Fig. 5). Confidence carries a depth-independent correctness signal (partial $r = 0.478 \approx$ raw). To our knowledge this is the first calibration audit in a single-cell ATAC modality; the closest prior work [41] applies conformal annotation to scRNA only, without a frozen-FM target, a coverage-gap-under-shift measurement, or depth deconfounding. (The load-bearing finding—whether *coverage* holds under shift—is a cal/test-exchangeability property computed identically for every method, robust to the cluster-A circular-label concern.)

We then audited two independent *real* scATAC FMs (Table II, Fig. 5B,C): ChromFound’s zero-shot embedding does *not* beat its own raw TF-IDF input (accuracy 0.834 vs. 0.850; ECE 0.094 vs. 0.088).

Atacformer [12] (a transformer encoder over an 890k-region genome universe; loader and official-code fidelity in §III-F) is a stronger embedding—accuracy 0.949, ECE 0.005—yet still does *not* beat the simple full-peak LSI baseline (0.968).

The two FMs confirm calibration tracks dimensionality, not FM-ness: the 192-dimensional Atacformer is as well-calibrated as a 50-dimensional SVD (ECE 0.005 vs. 0.007), whereas any 2048-dimensional representation (ChromFound or its raw input) is not (~ 0.09). Conformal coverage is robust for *both* FMs and baselines—no collapse under cross-sample (Atacformer gap -0.010) or cross-diagnosis Control $\rightarrow$ AD shift (gap -0.013), with a depth-independent confidence signal (partial $r = 0.491$) and negative controls at chance. So across two scATAC FMs the FM gives no accuracy gain over simple baselines, yet calibration/coverage stays robust to shift—the modality contrast we quantify on a common batch-shift ruler in cluster K (Fig. 6B).

Table II gives the full reliability profile (calibration, three coverage regimes, depth-deconfounded confidence, selective AURC, rare-type bias, negative controls) for all three representations: robustness is label-independent (negative controls collapse to chance or below—ChromFound’s 2048-d shuffle drops under it), the constructive counterpart to the scRNA

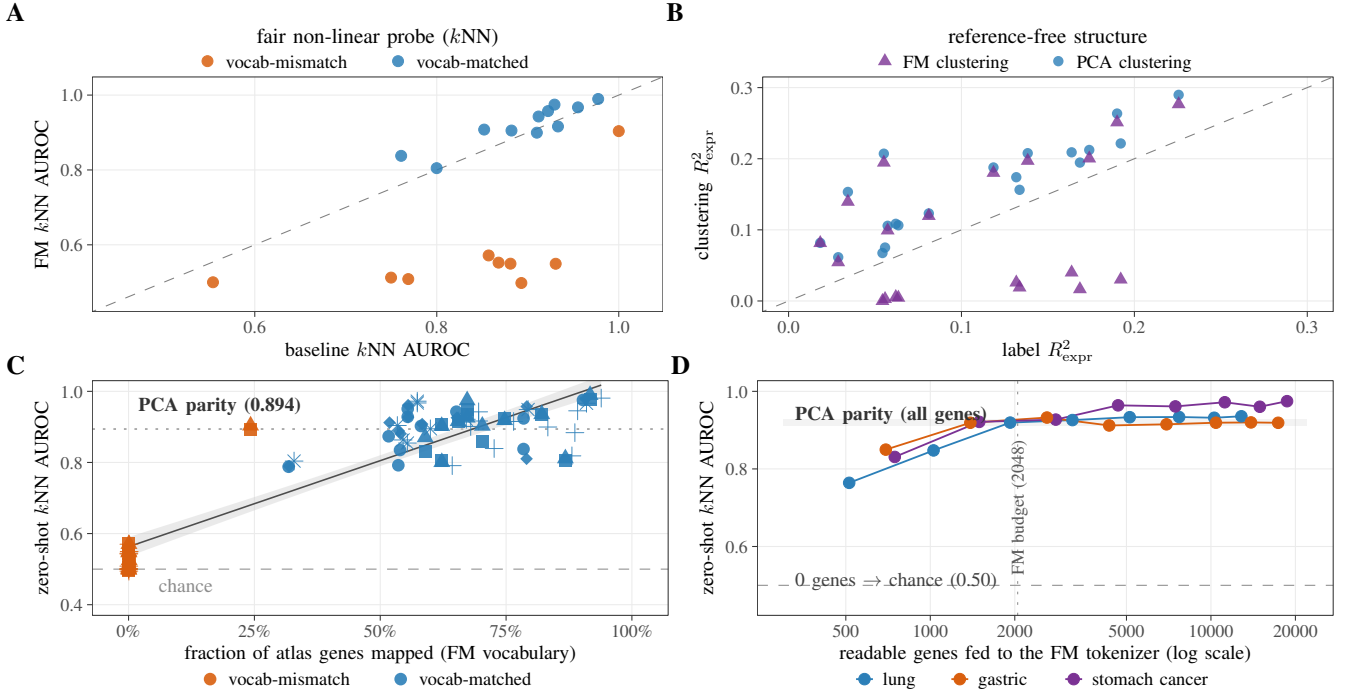


Fig. 3. *The corrected view and the vocabulary artifact (cluster J).* (A) Fair non-linear (kNN) probe, six FM arms over five families (best per atlas). On the 11 *vocabulary-matched* human atlases (blue) the FM sits on/near the diagonal (this best-of view carries a $+0.024$ max-selection edge; unbiased parity is the per-family test of Table III); the off-diagonal floor is the 8 cross-species mouse-symbol atlases (orange), where the tokenizer maps almost no genes ($kNN \approx 0.53$)—a tokenization handicap; the 9th (ENSG-only) sits near parity because its FM still maps $\sim 24\%$ of genes. (B) Reference-free R^2_{expr} : points above the diagonal mean the cell-type labels explain *less* HVG-expression variance than the PCA clustering (20/20 atlases; both clusterings 11/20)—the labels are a *coarser* partition than clustering recovers, corroborating the circularity. Together (A)–(B): with vocabulary mismatch and label circularity separated out, zero-shot scFMs *neither beat nor trail* PCA on coarse human cell types. (C) Vocabulary dose-response: x = fraction of atlas genes in the FM vocabulary, y = kNN -AUROC. Quality tracks coverage ($\rho = 0.83$, $r = 0.93$) but as a *threshold*—mismatch at $x \approx 0$, matched at $[0.32, 0.94]$, little between; the within-matched gradient is weak ($\rho \approx 0.34$ – 0.40). (D) Causal within-atlas ablation (three fixed atlases, only a fraction of readable genes kept, re-embedded; x log scale): AUROC flat at parity (grey = full-gene PCA) to ~ 1 – 2 k genes, then cliffs to 0.50 at zero—the cross-species “FM failures” are a tokenization cliff, not a representational deficit.

collapse in cluster K.

TABLE II

SCATAC COMBINED COMPARISON AND RELIABILITY PROFILE (CLUSTER I). **TOP** (MATCHED 20,002-CELL COMPARISON): NEITHER OF TWO INDEPENDENT SCATAC FMS BEATS THE SIMPLE FULL-PEAK LSI BASELINE; CALIBRATION (ECE) TRACKS *dimensionality*, NOT FM-VS-BASELINE (192-D ATACFORMER AS WELL-CALIBRATED AS THE 50-D SVD, 2048-D CHROMFOUND NOT); CONFORMAL COVERAGE ROBUST FOR ALL. **BOTTOM** (FULL 130,418-CELL PROFILE; PEAK-LSI IS THE FULL COHORT, THE TWO FMS THE MATCHED SUBSET): PEAK-LSI AND *both* FMS STAY WELL-CALIBRATED AND KEEP COVERAGE UNDER CROSS-SAMPLE AND CONTROL $\rightarrow$ AD SHIFT, WITH A DEPTH-DECONFOUNDED CONFIDENCE SIGNAL AND NEGATIVE CONTROLS AT/BELOW CHANCE—THE POSITIVE COUNTERPART TO THE SCRNA COLLAPSE (CLUSTER K); RARE-TYPE (PERICYTE/ENDOTHELIAL) OVER-REJECTION IS THE SHARED FAIRNESS COST. TOP/BOTTOM USE SEPARATE SPLITS, SO SHARED-BASELINE ESTIMATES DIFFER SLIGHTLY.

Representation	dim	accuracy	ECE	cross-sample coverage
ChromFound FM (faithful recipe)	2048	0.834	0.094	0.904
raw top-2048 log-TF-IDF (ChromFound's own input)	2048	0.850	0.088	0.909
Atacformer FM (full-universe, self-written)	192	0.949	0.005	0.910
LSI top-2048 $\rightarrow$ SVD-50 (matched peaks)	50	0.863	0.007	0.903
LSI full-peak $\rightarrow$ SVD-50 (reference)	50	0.968	0.011	0.914

	FM-free peak-LSI	ChromFound FM	Atacformer FM
cell-type accuracy	0.983	0.834	0.949
ECE	0.006	0.094	0.005
coverage, random	0.901	0.899	0.900
coverage, cross-sample	0.908	0.912	0.910
coverage, Control $\rightarrow$ Control (in-dist)	0.894	0.910	0.919
coverage, Control $\rightarrow$ AD	0.922	0.927	0.932
confidence partial- r (deconfounded)	0.478	0.438	0.491
AURC (selective)	0.001	0.039	0.006
rare-type (pericyte/endothelial) rejection	0.58	0.62	0.38
neg-control (label shuffle; chance 0.45)	0.45	0.24	0.45

F. Self-computed scRNA suite: representation (cluster J)

We re-derived the core claims on local, leakage-controlled scRNA atlases (frozen zero-shot embeddings; leave-one-batch-out readouts; deterministic seeds), each at its adversarially-verified confidence level (Sec. III-F).

On four atlases (lung-GEO, B-cells, cardiac-lymph, intestine; 4–26 donors) under leave-one-batch-out, scGPT does *not* dominate baselines: PCA or scVI [47] tops every atlas, scGPT is never best (Fig. 2C), is worst on lung in 4/4 folds with non-overlapping ranges ($scGPT \leq 0.961 < PCA \geq 0.979$), and a second FM (Geneformer) is weakest in 3/3 atlases. For batch *integration* the advantage is axis-specific: scGPT mixes batches better (higher iLISI in 3/3) but blurs cell types (worse cLISI; cardiac 5.33 vs. PCA 2.20), while scVI dominates the mixing/purity trade-off (Fig. 2D)—this reproduces the external cluster-A literature, not a universal FM-failure claim.

We then *expanded* the audit: *Data*, we grew the comparison from 3 to 20 raw-count atlases by transferring cell-type labels from curated (normalized) atlases onto raw-count twins via cell-barcode matching ($\geq 50\%$ overlap; typically 100%). *Baselines*, an enlarged panel (PCA+logreg, HVG+logreg, kNN , nearest-centroid, random-forest); the reliability sweep spans 24 atlases (cluster K). *Models*, six FM arms across five families run zero-shot via self-written loaders (§III-F); the primary embedders are Geneformer-V2-104M and scGPT-human (scGPT

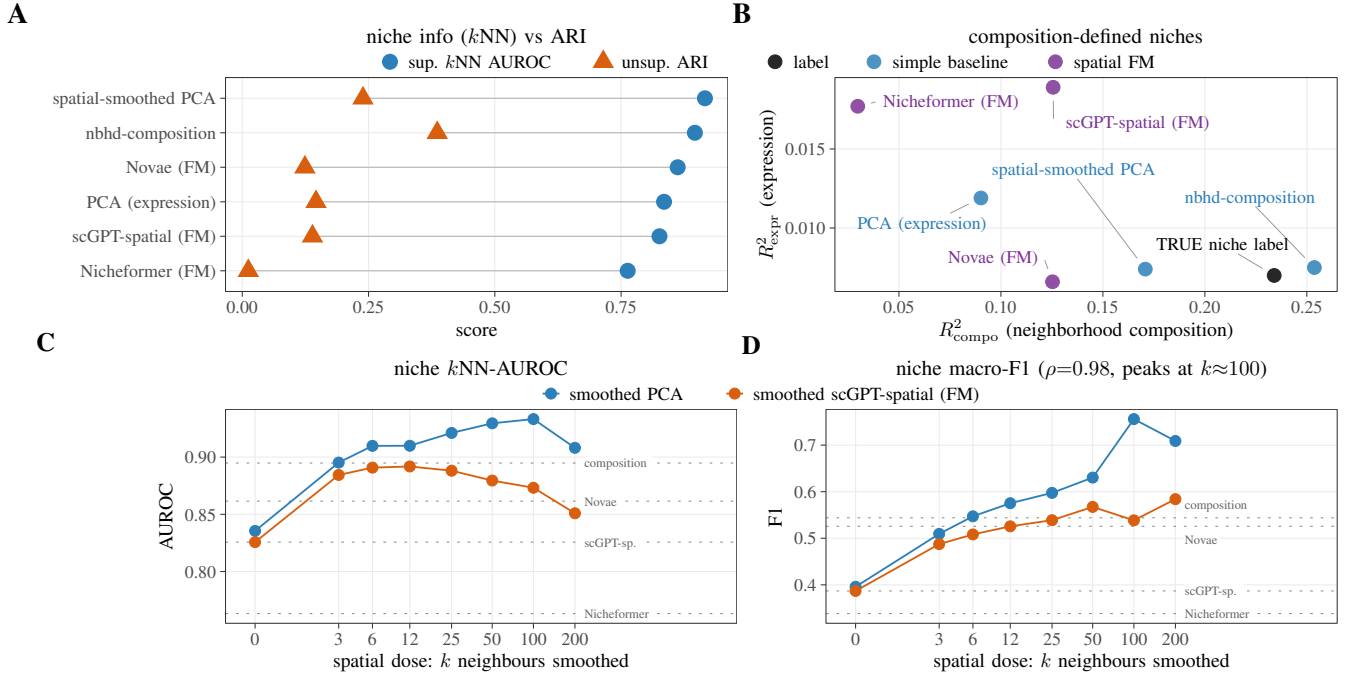


Fig. 4. Cluster-H fair re-check and spatial dose-response (niche-ID). (A) unsupervised ARI-vs-label (orange) says the spatial FMs fail, but a supervised k NN niche-probe (blue) shows all three carry niche information (AUROC 0.76–0.86); the residual gap to spatial baselines is small. (B) the niche labels are composition-defined (R^2_{compo} 0.234, near-zero single-cell R^2_{expr} 0.007), so the composition baseline winning ARI is near-tautological—the same circularity as cluster J. (C–D) spatial dose-response: per-cell representations are progressively smoothed over k spatial neighbours (x log scale, $k=0$ per-cell), scored by supervised niche prediction (spatially-blocked CV): (C) k NN-AUROC, (D) macro-F1. Smoothing per-cell PCA (blue) lifts prediction monotonically (F1 $\rho = 0.98$) past all three row FMs and the composition baseline by $k \approx 25$ –50; the same dose helps a smoothed FM (orange). The residual “spatial-FM gap” is thus the spatial-aggregation axis—one operation a per-cell model lacks—not a representational deficit.

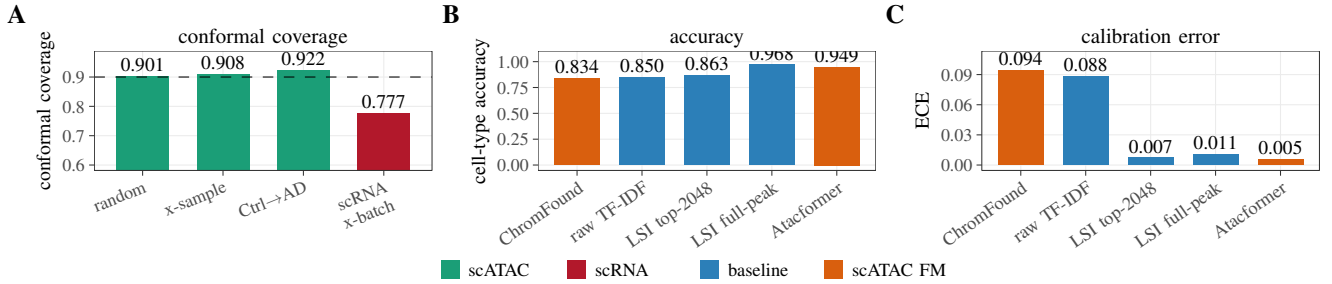


Fig. 5. *scATAC* (cluster I). (A) Conformational coverage holds under cross-sample and cross-disease shift, whereas the scRNA scGPT regime collapses. (B,C) Matched comparison of two scATAC FMs against simple LSI baselines: neither FM beats full-peak LSI on accuracy, and the ECE difference is a representation-dimensionality effect.

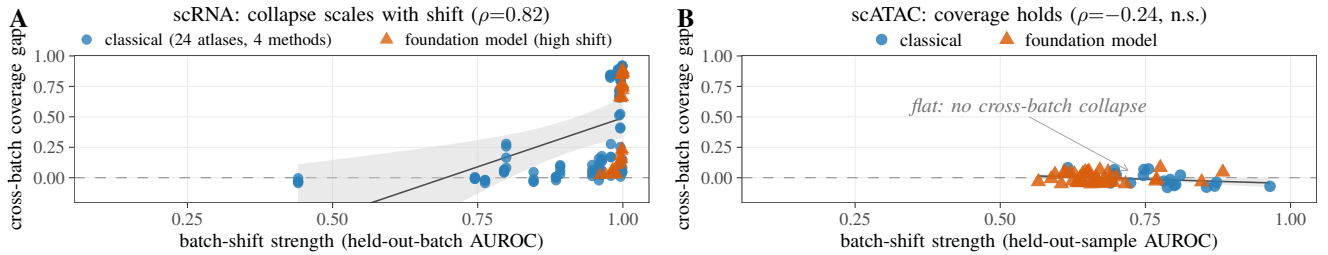


Fig. 6. *Batch-shift dose-response and modality contrast, on one ruler.* x = batch-shift strength (cross-validated AUROC for held-out-batch/sample membership; 0.5 exchangeable, 1.0 separable); y = cross-batch conformational coverage gap; axes shared. (A) scRNA: blue = the 24-atlas $\times$ 4-classical-method sweep (shift 0.13–1.0; fitted curve and $\rho = 0.82$ are PCA+logreg); three FM families (orange, vocabulary-matched atlases) reach only the high-shift end (≥ 0.96) and collapse by the same amount as classical (+0.48 vs. +0.49)—FMs do not repair it. (B) scATAC (same axes): peak-LSI plus two FMs (ChromFound, Atacformer) give a *flat* curve—gap near zero throughout ($\rho = -0.24$, n.s.; mean -0.005 , max $+0.08$). The scRNA collapse also holds for the ECE gap ($\rho = 0.76$) and a classifier-free centroid-distance measure ($\rho = 0.58$ –0.60). Coverage collapse is a graded property of the scRNA cross-batch regime—shared by every method there, absent in scATAC.

reproduces the official cached embedding); four added later—Geneformer-V2-316M, scFoundation, CellPLM, UCE—enter the fair check below.

A naive scoring—cross-batch macro-AUROC of a *linear* probe against the labels—lets a baseline “match or beat” at least one FM in 16/20 atlases (*both* in 14/20; Fig. 7), largely a metric artifact the fair re-check (Fig. 3) corrects.

G. Self-computed scRNA suite: the fair re-check and parity (cluster J)

The re-check corrects it on two fronts. First, a *non-linear* probe (k NN) removes the bias a linear probe gives PCA: under k NN the best FM *ties or beats* the best baseline on all 11 vocabulary-matched human atlases (tie band $|\Delta\text{AUROC}| < 0.02$; e.g. stomach 0.97 vs. 0.93, breast-metastasis 0.91 vs. 0.85). Second, a *reference-free* structure metric—HVG expression variance explained by each method’s unsupervised partition (R^2_{expr} , no labels)—finds FM and PCA partitions essentially equal on non-collapse atlases (0.163 vs. 0.172), while labels explain *less* expression variance than PCA clustering in 20/20 atlases (and than *both* clusterings in 11/20; Table III).

Nor is this a 2000-HVG-space artifact favoring PCA: the labels-below-PCA count stays 20/20 recomputing R^2_{expr} at 1000–4000 HVGs and over *all* genes. A null control addresses the opposite objection (that variance-minimising K -means is favoured over labels): against a size-matched *random* partition the labels explain $\sim 24\times$ more HVG variance (17/17 atlases), yet sit below the K -means ceiling—not noise, but a *coarser* partition than clustering recovers.

Clustering the FM embedding (not PCA) cross-checks this: labels fall below the FM clustering only on matched atlases where the FM is functional (9/17), since on cross-species/gene-ID atlases the FM embedding is near-degenerate ($R^2_{\text{expr}} \approx 0.004$) and cannot out-explain even the labels (Table III). Thus the AUROC race reproduces this coarser cut more than it captures the finer structure clustering finds—where FM and baseline tie.

Labels under-explain variance partly by construction, so circularity rests not on the 20/20 count but on probe asymmetry (a linear probe in the labels’ own space flatters PCA; k NN removes the gap), corroborated by a direct test (synthetic self-clustered labels, illustrative): each representation scores higher against labels from its *own* space (representation $\times$ label-source interaction $+0.055$, 95% CI $[+0.039, +0.072]$, 9/9 atlases)—the metric measures proximity to the label-generating method, the same coarseness inflating ARI/scIB in the cluster-A literature. The second artifact is the *location* of FM failure: our earlier read attributed three “human” collapses to fine subtypes, but all three carry *mouse* gene symbols ($Xkr4$, Gm -, Rik -genes), so their collapse is the same cross-species token-map failure. Re-classifying all 20 atlases by actual vocabulary leaves 11 vocabulary-matched human, 8 cross-species (mouse-symbol), and 1 gene-ID-mismatched (ENSG-only, degenerate for the tokenizer, trivially separable for PCA; the 9th vocab-mismatch point in Fig. 3A).

On the 8 cross-species atlases the human-symbol FMs (scGPT, Geneformer-V2 104M/316M) drop to k NN ≈ 0.53 (vs.

PCA 0.81, which needs no gene map)—a disclosed tokenization handicap benchmarks rarely separate from representation. On the 11 matched atlases all five families are at parity with PCA on both the k NN probe (family means straddling PCA’s 0.894 within ≤ 0.024) and R^2_{expr} , each tying or beating PCA in 7–10 of 11 (7 of 10 for UCE, which drops retina; Table III). The lone clean atlas where most FMs trail is bone marrow (0.83–0.88 vs. 0.91).

“Parity” is a null-equivalence claim, tested as one, not inferred from close means. For each family we bootstrap the paired per-atlas difference from PCA (k NN-AUROC) over 11 atlases against a pre-stated ± 0.02 margin (the tie band). No family differs from PCA (Wilcoxon $p \geq 0.27$ for all six—no advantage *and* no deficit); pooled, the difference is $+0.0025$ (naive 95% CI $[-0.006, +0.010]$). Since the six arms share the 11 atlases, the correct atlas-cluster bootstrap roughly doubles the width to $[-0.013, +0.020]$; the equivalence is positive but *marginal*—the cluster CI’s upper edge ($+0.0199$) sits *at* the ± 0.02 margin, within Monte-Carlo error, a boundary non-rejection, not resting on the TOST alone. One asymmetry matters: the self-written loaders are pessimistic-for-the-FM (a better-tuned official pipeline can only *raise* an FM’s score), so the *no-deficit* half is firm while *no-advantage* is conservative. Per arm, parity is certified for scGPT, Geneformer-V2-316M, and UCE (loaders verified against official code, §III-F); Geneformer-V2-104M, scFoundation, and CellPLM have per-arm CIs exceeding ± 0.02 —underpowered on 11 atlases (point estimates within ± 0.011 of PCA).

We also avoid the “best-FM-per-atlas” summary, whose $+0.024$ edge over PCA is a max-selection artifact; the per-family test is unbiased. Parity survives three robustness checks: probe neighbourhoods ($k \in \{5, 15, 30, 50\}$), twice as many clusters as labels (R^2_{expr} gap 9/9 atlases at $K=\text{NC}$ and 2NC), and two *non-distance* probes (logistic, 128-unit MLP; both straddling zero). The corrected verdict—zero-shot scFMs neither beat nor trail simple baselines on vocabulary-matched coarse cell types—replaces the “baselines decisively win” of our earlier single-metric read (which averaged the cross-species handicap into the “human” mean). To test capacity vs. architecture we added Geneformer-V2-316M ($3\times$ parameters) and three families (scFoundation, CellPLM, UCE): tripling parameters does not improve on the 104M sibling, and all three land at the same parity—“not governed by model scale or architecture family” (Table III; see also [30]).

H. Self-computed scRNA suite: the vocabulary-coverage artifact (cluster J)

A vocabulary dose–response makes this artifact mechanistic: coverage (the fraction of atlas genes in each FM’s vocabulary) plotted continuously against zero-shot k NN-AUROC (Fig. 3C) shows quality rising mostly as a *cliff*—atlases split into a mismatch group mapping essentially no genes (8 cross-species atlases, plus the ENSG atlas for scGPT; fraction ≈ 0) and a matched group at 32–94%, almost nothing between except the ENSG atlas under Geneformer-V2 ($\sim 24\%$, at parity).

The full-span association is strong (three universal arms, scGPT and Geneformer-V2 104M/316M; 60 points: Spearman

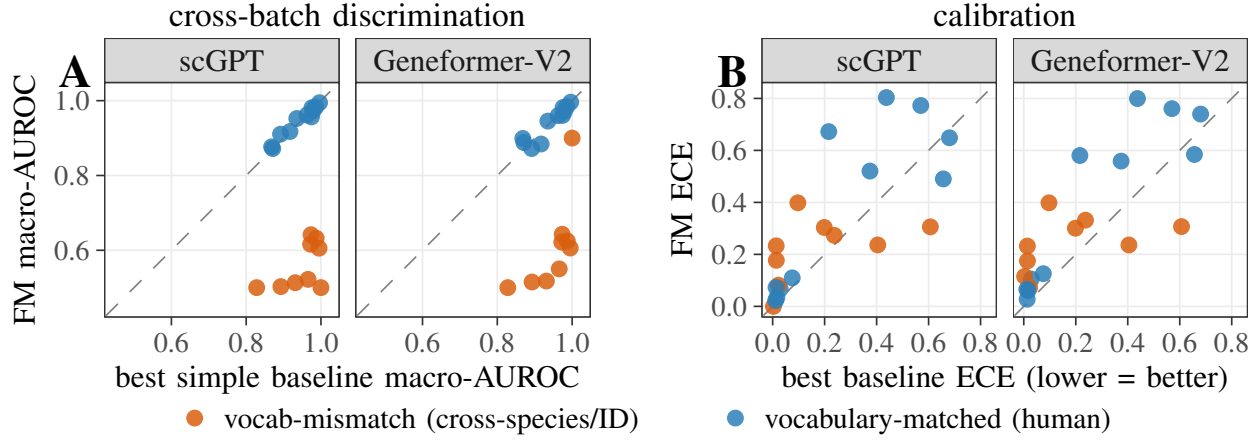


Fig. 7. *The metric-artifact view (cluster J): two zero-shot FMs (scGPT + Geneformer-V2) vs. the best simple baseline across 20 raw-count atlases (3 native + 17 unlocked by barcode label-transfer). (A)* Cross-batch discrimination, scored by a *linear-probe* AUROC against the labels: the apparent baseline “win” is concentrated in the orange *vocab-mismatch* atlases (8 cross-species mouse-symbol + 1 ENSG-only, where the human-symbol FM tokenizer maps almost no genes); on the 11 blue vocabulary-matched atlases the points hug the diagonal, and the fair re-check (Fig. 3) removes the residual gap as circularity and probe bias. *(B)* The matched calibration view (FM ECE vs. best-baseline ECE, lower is better): the FMs sit mostly above the diagonal (worse-calibrated out-of-batch), the reliability counterpart that cluster K quantifies and partly repairs.

$\rho = 0.83$, Pearson $r = 0.93$, $p < 10^{-26}$; reference-free, R_{expr}^2 , $\rho = 0.68$, $p < 10^{-8}$) but dominated by the mismatch/matched gap—within the matched group the residual gradient is weak ($\rho \approx 0.34$ by coverage fraction, 0.40 pooled across families, $\rho \approx 0$ by expression mass). It is thus a vocabulary *threshold*, not a dose: below it, atlases sit near chance (mean AUROC 0.53); above, they plateau at PCA parity ($0.90 \approx \text{PCA's } 0.894$). Cross-species/gene-ID “FM failures” are a tokenization cliff, not a representation failure.

We confirm this causally, within a single atlas (Fig. 3D): on three fixed matched atlases we keep only a fraction of tokenizer-readable genes (renaming the rest to map to nothing) and re-embed, removing the cross-atlas biology confound; PCA on the identically renamed matrix is unchanged (a built-in tokenizer-free control). The result: a clean *plateau-then-cliff*— $k\text{NN-AUROC}$ holds parity (0.92–0.97) down to $\sim 1\text{--}2\text{k}$ readable genes, then collapses to exactly chance (0.500) at zero (Spearman frac-vs-AUROC 0.52–0.95). Coverage thus *causes* the apparent quality: the FM needs only $\sim$ its top-gene budget to match PCA, and the cross-species atlases (and ENSG for scGPT) sit at ~ 0 , below the cliff (ENSG excepted: Geneformer-V2 maps $\sim 24\%$, at parity). The cliff is *FM-general*: a second family (scGPT, value-binning not rank-value) gives the same result on all three atlases (Spearman 0.73–0.98, parity to $\sim 1\text{--}2\text{k}$ genes, chance at zero).

I. Self-computed scRNA suite: calibration and reliability (cluster K)

Frozen scGPT cross-batch *discrimination* transfers (LOBO macro-AUROC ~ 0.95 ; shuffled-label control 0.48), but *calibration* does not: cross-batch ECE is 0.17–0.23, and higher readout capacity (linear $\rightarrow$ MLP-256) worsens out-of-batch ECE in 4/4 atlases with no in-distribution gain (Fig. 8A,B); temperature scaling [40] closes only $\sim 1/3$ of the gap (0.176 $\rightarrow$ 0.126). Split-conformal coverage is *unstable across batches*: positive in 3/3 non-lung atlases (0.012–0.075), reach-

ing 0.123 [0.115, 0.131] on lung under a binomial-count depth ablation (down-sample-and-re-embed; 0.123 $\rightarrow$ 0.169 as depth drops)—the *mild* end of a larger collapse for classical baselines. After deconfounding by log-depth, log-gene count, and batch (Frisch–Waugh–Lovell), confidence retains a residual benchmark-correctness signal (partial r 0.25–0.40): a cross-batch/exchangeability instability [14], not a leakage or biological-validity claim. The closest prior work [41] targets neither a frozen FM nor a coverage-gap-under-shift measure.

Expanding to 24 atlases reframes the finding: the gap is large for *simple* baselines too—mean +0.35 [0.21, 0.51] for PCA+logreg, +0.38 random-forest, +0.37 $k\text{NN}$, positive in 20/24 atlases (max 0.92) (Fig. 8C)—with the scGPT lung value (0.123) at the *milder* end. The defensible claim: cross-batch conformal coverage collapse is a general exchangeability failure under batch shift [14], afflicting classical and foundation models alike (often more), not an scFM-unique pathology.

The robust contribution is the *modality contrast*: ubiquitous in scRNA cross-batch annotation (20/24 atlases, all methods), absent in scATAC coarse-type annotation under sample/disease shift (cluster I). Nor is the collapse fatal: selective abstention recovers accuracy across the 24 atlases (accuracy at 80% coverage exceeds full-coverage for every baseline), and temperature scaling lowers but does not erase cross-batch ECE (Fig. 8D,E)—partial, deployable mitigations, not a cure.

Two robustness checks back this (Fig. 8F,G): the readout-capacity $\rightarrow$ ECE effect reproduces on a second FM (Geneformer-V1-10M, pooled leave-one-batch-out)—MLP-256 again worst-calibrated, discrimination flat, so over-capacity miscalibration is a readout property, not one model; and on the lung scGPT probe the reject option is effective (selective AURC 0.084 vs. 0.259 random) and *fair to rare cell types*, retained *more* than common ones at 80% coverage (0.90 vs. 0.79), with rejected cells disproportionately errors (rejected-rare accuracy 0.54 vs. retained-rare 0.91).

A batch-shift dose-response quantifies the exchangeability-

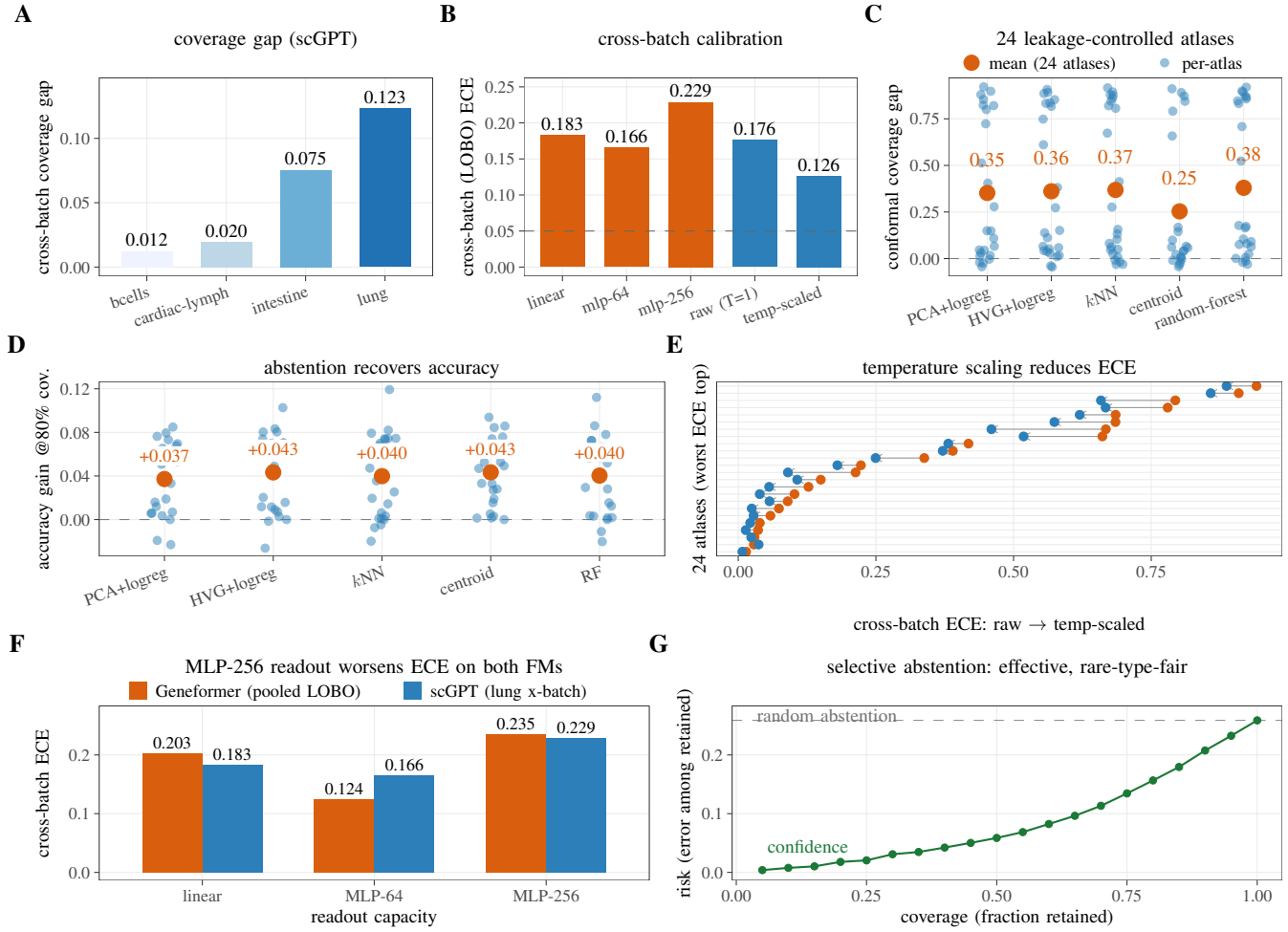


Fig. 8. *Cluster-K reliability: cross-batch collapse (A–C) and its recovery/robustness (D–G).* (A) frozen scGPT cross-batch conformal coverage is unstable (gap > 0 in all four atlases; 0.012–0.075, and 0.123 on lung under the depth ablation). (B) discrimination transfers but calibration does not—the highest-capacity readout (MLP-256) worsens out-of-batch ECE, temperature scaling closes only $\sim 1/3$ of the gap. (C) 24-atlas reframe: the collapse is large and pervasive for simple baselines too (blue = per-atlas, orange = across-atlas mean; positive in 20/24)—a general exchangeability failure, not an FM defect. (D) selective abstention recovers accuracy (80%-coverage accuracy exceeds full-coverage for every baseline). (E) temperature scaling lowers but does not erase cross-batch ECE—partial remedies, not a cure. (F) the readout-capacity $\rightarrow$ ECE effect is FM-general (worst at MLP-256, best at MLP-64 for both scGPT and Geneformer-V1-10M): an over-capacity effect. (G) the lung scGPT reject option beats random abstention (AURC 0.084 vs. 0.259) and is *fair to rare types*—retained more than common types (0.90 vs. 0.79), rejected rare cells disproportionately errors (0.54 vs. 0.91).

failure claim: a cross-validated classifier in the same PCA space predicts held-out-batch membership, its AUROC (0.5 exchangeable, 1.0 separable) the violation strength. Across the 24 atlases the gap is a graded function of it—Spearman $\rho = 0.82$ (PCA+logreg; $p < 10^{-5}$)—surviving every check (Fig. 6): pooled PCA baselines $\rho = 0.75$, HVG+logreg $\rho = 0.80$, the ECE gap $\rho = 0.76$, a classifier-free centroid distance $\rho = 0.58$ –0.60. It is predominantly *exchangeability*, not lost signal: at high shift the classifier still ranks cells well (AUROC 0.92) while coverage drops 0.49 ($\sim 8\times$ larger)—miscoverage from score-distribution shift, not an unlearnable task.

Foundation models collapse on the same curve. Three FM families (scGPT, Geneformer-V2-104M, scFoundation) as representation: on vocabulary-matched atlases (high-shift in both PCA and FM space, tracking within 0.011, Pearson 0.97) the FMs reach only the *high-shift end* (≥ 0.96 ; Fig. 6A) and collapse by the same amount as classical baselines (+0.48 vs. +0.49). Near-exchangeable batches (shift < 0.9) show

essentially no collapse (+0.03); strongly non-exchangeable ones an order of magnitude more (+0.5): collapse tracks data regime, not representation.

The same ruler quantifies the modality contrast: the identical protocol on the scATAC atlas (cluster I; peak-LSI, Chrom-Found, Atacformer) gives the opposite curve (Fig. 6B): flat in shift ($\rho = -0.24$, *n.s.*), near zero throughout (mean -0.005 , max $+0.08$), never collapsing even at peak separability. The slopes differ sharply—scRNA $+1.61$ vs. scATAC -0.14 (interaction $p = 1.6 \times 10^{-5}$; label-permutation $p = 3 \times 10^{-4}$)—but read as *within-dataset descriptive*: the scATAC arm is a single atlas (GSE174367), so a modality claim needs ≥ 2 datasets (§Limitations). The contrast is a *modality $\times$ regime* property, not an FM one.

J. Self-computed scRNA suite: perturbation (cluster L)

On Norman-2019 [48] double perturbations, an additive linear prior (control $+\delta_A + \delta_B$) outperforms the deep GNN GEARS [28]. A *label-fair, full-gene* comparison (no oracle

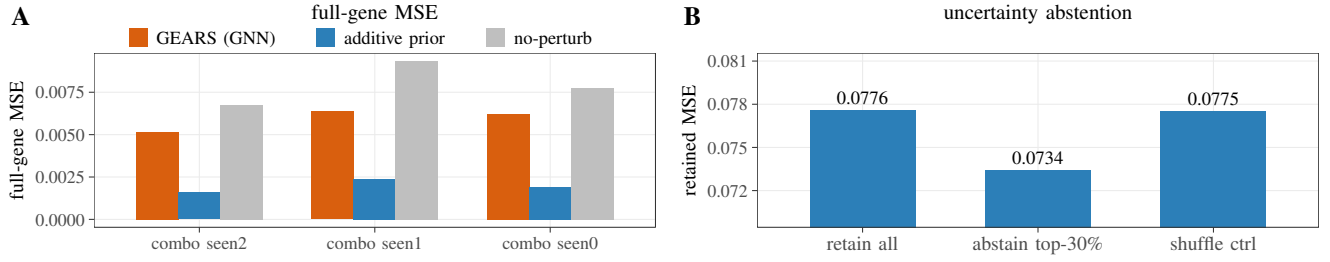


Fig. 9. Cluster L (perturbation). (A) a label-fair, full-gene comparison (no oracle gene selection) on Norman-2019—the additive prior beats the deep GNN GEARs on all three combo strata ($\sim 3\times$ lower MSE; 2.7–3.3 $\times$), upgrading cluster L from exploratory to supported. (B) GEARs’ own predicted uncertainty gives a genuine but modest selective-prediction gain (abstaining on the top-30% most-uncertain cells lowers retained MSE; partial Spearman 0.65 after effect-magnitude deconfounding). The panel-B y -axis is truncated (bar labels give absolute MSE); the reduction is $\sim 5\%$, small in absolute terms.

TABLE III

PER-FAMILY PARITY SUMMARY AND PER-ATLAS NULL-PARTITION DETAIL (CLUSTER J). **TOP:** MULTI-FM CHECK ON THE 11 VOCABULARY-MATCHED HUMAN ATLASES UNDER THE FAIR k NN PROBE AND REFERENCE-FREE R^2_{expr} . FIVE ARCHITECTURALLY INDEPENDENT FM FAMILIES SPAN MEAN k NN 0.886–0.910 around PCA’s 0.894—PARITY, NONE DECISIVELY BEATING OR TRAILING PCA, WHILE HVG ALONE IS CLEARLY WORSE; TRIPLING PARAMETERS (GENEFORMER 104M $\rightarrow$ 316M) BUYS NOTHING. “TIES/BEATS” COUNTS ATLASES WHERE THE FM IS WITHIN 0.02 OF, OR ABOVE, PCA. THE 9 VOCAB-MISMATCH ATLASES ARE ANALYSED SEPARATELY (FIG. 3). **BOTTOM:** NULL-PARTITION CONTROL (HELD-OUT BATCH OF EACH RAW-COUNT ATLAS)—FRACTION OF HVG-EXPRESSION VARIANCE EXPLAINED BY A SIZE-MATCHED *random* PARTITION, THE TRUE *labels*, THE PCA K -MEANS CLUSTERING, AND THE BEST FM CLUSTERING. LABELS BEAT THE RANDOM NULL IN 17/17 (NOT NOISE) YET FALL BELOW THE PCA CLUSTERING IN 17/17 (THE CLUSTER-J FINDING, WITH A NULL FLOOR), AND BELOW THE FM CLUSTERING IN ONLY 9/17—ON CROSS-SPECIES/GENE-ID ATLASES THE FM EMBEDDING IS ITSELF NEAR-DEGENERATE ($R^2_{\text{expr}} \approx 0$), THE FIG. 3C,D TOKENIZATION FAILURE READ THROUGH CLUSTERING. COVERS 17 OF 20 ATLAS (THOSE ADMITTING A SIZE-MATCHED RANDOM PARTITION).

representation	mean k NN AUROC	mean R^2_{expr}	ties/beats PCA
best simple baseline (PCA)	0.894	0.219	11/11
scGPT	0.894	0.205	10/11
Geneformer-V2-104M	0.905	0.219	9/11
Geneformer-V2-316M	0.894	0.225	9/11
scFoundation	0.904	0.215	8/11
CellPLM	0.886	0.168	7/11
UCE [†]	0.910	0.210	7/10
HVG (2000 genes)	0.750	0.232	0/11

[†]UCE applies its own min-count cell filter; one atlas (retina, 5 cells dropped) fell out of position-alignment and is omitted, so UCE is scored on 10/11.

atlas	#labels	random	labels	PCA-clust	FM-clust
ad_hm	15	0.001	0.226	0.306	0.285
breast_hm	18	0.004	0.192	0.235	0.026
ca_cancer	17	0.003	0.190	0.266	0.252
gastric	20	0.006	0.174	0.214	0.201
hesc_hspc_cd8	15	0.003	0.169	0.196	0.014
astrocytes_sci	20	0.007	0.164	0.211	0.040
lung_adre	22	0.011	0.138	0.209	0.193
tcell_cancer	16	0.005	0.132	0.175	0.026
bm_all	13	0.006	0.118	0.189	0.179
progastin	21	0.003	0.064	0.122	0.004
urine	24	0.003	0.062	0.110	0.005
lsk_batch	11	0.002	0.056	0.076	0.003
breast_hm2	18	0.007	0.055	0.212	0.195
lps_mm	4	0.001	0.054	0.071	0.000
stomach_cancer	17	0.005	0.034	0.157	0.138
aml_pbmc	19	0.005	0.029	0.063	0.053
breast_metastasis	26	0.006	0.018	0.088	0.079
mean		0.005	0.110	0.171	0.099

gene selection)—the test required before asserting GEARs is beaten—confirms it: additive beats GEARs on *all three* combo strata (both-singles-seen full-gene MSE 0.0016 vs. 0.0051, $\sim 3.2\times$; Fig. 9), so cluster L is *supported*, consistent with cluster B and the external linear-baseline result [24]. Separately, GEARs’ own predicted uncertainty carries a genuine but modest selective-prediction signal surviving effect-magnitude deconfounding (partial Spearman 0.65; abstaining on the top-30% most-uncertain cells lowers retained MSE 0.0776 $\rightarrow$ 0.0734 vs. a shuffled-uncertainty control 0.0775).

V. DISCUSSION

Three conclusions are robust across eleven task clusters (Table I), external and self-computed. First, zero-shot scFMs give no advantage—and no deficit—over simple baselines on vocabulary-matched coarse cell types: parity on representation/annotation/integration and spatial niche-ID, real-but-metric-contingent for perturbation, no gain over their own input for a real scATAC FM. “No advantage” not “base-lines win”: the apparent margin is two artifacts our re-check isolates—label circularity (a non-linear probe and label-free R^2_{expr} tie FM and baseline; labels explain less variance than PCA in 20/20 atlases, Fig. 3) and a vocabulary-coverage threshold (FM shortfalls confined to cross-species/gene-ID atlases where the tokenizer reads almost no genes). Three depth probes (Table IV) make each a measured dose–response.

Second, the much-warned cross-batch calibration/coverage collapse is a *general* exchangeability failure of the scRNA regime, not an FM defect: it hits classical baselines as much as frozen embeddings (often more), grows with held-out-batch non-exchangeability (Fig. 6), and is *absent* in scATAC under sample/disease shift—what an FM would repair belongs to the modality and regime, not the model. Reliability is recoverable where data are exchangeable (scATAC, Table II) and partially repairable where not (abstention/recalibration, Fig. 8D,E); even where pretraining helps—FM batch mixing—it costs cell-type purity, which scVI avoids (Fig. 2D).

Third, which method “wins” is governed by task and evaluation choice, not model scale or architecture—visible in perturbation (metric family), scATAC calibration (dimensionality), and the probe/label circularity above. Tripling parameters (Geneformer 104M $\rightarrow$ 316M) and swapping three independent families (scFoundation, CellPLM, UCE) leave the result unchanged (Table III); pretraining’s value, where it

TABLE IV

THE THREE MECHANISTIC DEPTH PROBES: EACH CONVERTS A BINARY OR QUALITATIVE CLAIM INTO A MEASURED DOSE-RESPONSE, AND EACH IS COMPUTED WITH THE METHOD-CLASS THE CLAIM CONCERNS (FOUNDATION MODELS) ON THE CURVE. “RESULT” GIVES THE HEADLINE; FULL STATISTICS ARE IN §III-F AND THE CITED FIGURES.

depth probe	scientific question	driver (x)	outcome (y)	result
#1 vocabulary (scRNA repr.)	apparent FM failure: representational or tokenization?	mapped-gene fraction	k NN-AUROC	<i>threshold</i> ($\rho=0.83$); causal within-atlas ablation is flat at parity then cliffs to chance below ~ 1 –2k genes; FM-general (scGPT & Geneformer)
#2 batch-shift (scRNA reliab.)	coverage collapse: FM defect or general?	held-out-batch separability	coverage gap (& ECE gap)	<i>general</i> ($\rho=0.82$); method- and model-general (FMs on the same curve); predominantly exchangeability, not task difficulty; scATAC flat ($\rho=-0.24$)
#3 spatial context (niche-ID)	residual gap: representational or a missing operation?	spatial smoothing k	niche k NN-AUROC / F1	<i>aggregation axis</i> : spatial-smoothing per-cell PCA overtakes all three spatial FMs (F1 $\rho=0.98$); the FMs simply are not given the dose

appears, is fine-tuned or supervised-downstream, not zero-shot representation.

The headline is methodological: fair, reference-free re-scoring shows standard label-matching leaderboards do not adjudicate what they claim. The backbone is our own experiments (scRNA reliability suite over up to 24 atlases, the first spatial-FM head-to-head, the first scATAC audit), with the meta-analysis as supporting context.

We state these findings provisionally (one practitioner’s results on tractable subsets with self-written loaders; several clusters resting on few studies; caveats below), concluding neither that foundation models are useless nor that baselines are vindicated—only that current rules cannot adjudicate the question, and a fairer, reference-free test is now available.

Implications

For practitioners: (1) *Choose representations by task, not pedigree*—for zero-shot coarse-type annotation, integration, and niche-ID a cheap classical pipeline (PCA, scVI, neighbourhood composition, spatial smoothing) matches an FM at far lower cost; adopt an FM only where a concrete, typically fine-tuned, gain is shown. (2) *Do not trust a single label-matching score*—labels are derived from the expression space, so ARI/NMI/scIB/label-AUROC can rank the annotation pipeline, not the biology; pair any leaderboard with a non-linear probe and a reference-free metric, and argue “no difference” by equivalence test. (3) *Treat cross-batch uncertainty as unreliable by default*—conformal coverage collapses under batch shift for classical and foundation representations alike, so abstention/recalibration are necessary.

Methodologically, the move to carry forward is from binary verdicts to *dose-response with the criticized method class on the curve*: tying a failure to a manipulable driver converts a rank into a mechanism and, as the vocabulary ablation shows, can flip a “representational deficit” into a removable, causal artifact.

a) *Generality beyond genomics.*: Nothing in the protocol is biological: it takes a frozen encoder [2], [1], labels of contested provenance, and a shift covariate, and returns a fair-evaluation verdict plus reliability audit. Its three confounds recur wherever pretrained representations are ranked—

label circularity (“ground truth” derived in the graded space: weakly-supervised vision, LLM-judged text), fixed-vocabulary coverage, and broken exchangeability under covariate shift, undermining split-conformal coverage and the calibration temperature scaling repairs [38], [39], [14], [40]. Porting needs only naming the dose and a label-independent structure metric; a leaderboard should certify it measured a representation, not its own rules.

Limitations

Scope is strictly zero-shot. The pooled corpus is dominated in raw count by one study (mitigated by equal-weight estimation), cluster B rests on seven (several small) studies, and—assembled by a literature search—is subject to selection/publication bias, a further reason the meta-analysis is supporting context, not an unbiased estimate.

Self-computed *generality* is bounded by dataset count: the spatial, scATAC, and perturbation arms each rest on a *single* dataset (mechanism demonstrations, not multi-dataset establishments), while only the scRNA suite spans many atlases (up to 24). ChromFound ran on a disclosed top-2048-peak/20k-cell subset (a plausible but unverified lower bound).

The matched-vocabulary equivalence rests on 11 atlases and certifies 3 of 6 arms (the rest consistent with, but underpowered for, parity); its matched/mismatch split was revised after a gene-naming audit, and the pre-audit set (mis-filing three mouse-symbol atlases as human) turns parity into an apparent -0.041 deficit, so the revision removed an anti-FM artifact rather than engineering parity. The 20-atlas FM arm relies on barcode label-transfer onto raw-count twins ($\geq 50\%$ overlap), and the expanded panels omit scVI.

Several “ground-truth” labels are clustering-derived and thus partially circular for PCA; we correct with a non-linear probe and a reference-free R^2_{expr} whose HVG-space bias we bound. Reflexively, this re-check is itself evaluation choices—a k NN probe, a ± 0.02 margin, 11 atlases—so by our own third finding it is not the uniquely correct arbiter; we offer it as falsifiable, not final.

Finally, the FM loaders are faithful-but-not-official lower bounds—except scGPT and Geneformer-V2, which reproduce the official embeddings exactly (§III-F)—so five-family parity

rests partly on others' code. Integration-metric critiques (cluster D) caveat cluster A's scIB inputs, and cluster L rests on a label-fair full-gene comparison (the second-FM reproduction qualified, §III-F).

DATA AND CODE AVAILABILITY

Code, normalized comparison tables, self-computed audit outputs, and figure scripts are archived on Zenodo under a versioned DOI ([10.5281/zenodo.21071826](https://doi.org/10.5281/zenodo.21071826)), organized by cluster; curated code and a reusable protocol specification are at github.com/PeterPonyu/frozen-fm-eval. Every stage is a standalone deterministic script (fixed seeds); external per-comparison tables are reused from each study's public release, not redistributed (metric values only, ~0.3 GB).

Primary data are public: the snATAC atlas of Morabito *et al.* [46] (GEO GSE174367); the CosMx lymph-node niche dataset of Wang *et al.* [43] (lymph_node_niche_annotated.h5ad, <https://github.com/WYXNICK/spatial-niche-benchmark>); and Norman *et al.* 2019 Perturb-seq (GEO GSE133344) [48], the GEARS-distributed preprocessed AnnData; all model weights come from each model's public repository.

The 24 leakage-controlled scRNA atlases (clusters J, K) and the four primary leave-one-batch-out atlases (Figs. 2C, 8A,B; lung GEO GSE130148, the rest in atlas_manifest.json) are public GEO/CELLxGENE releases; the full accession mapping and 95-atlas inventory are in the Zenodo archive. Analyses run in a single GPU environment (PyTorch, scanpy, scikit-learn, two conformal libraries [49], [50]).

AUTHOR CONTRIBUTIONS

The analysis, audits, software, figures, and manuscript were produced as a single working effort, with independent adversarial verification as a separate pass that corrected an initial scATAC-FM overclaim into the matched-control statement reported here.

SCOPE AND ETHICS

All claims are restricted to the zero-shot / out-of-the-box regime and benchmark reliability; no clinical, diagnostic, or biological-discovery claims are made. Fine-tuned or supervised-downstream models are out of scope and can reverse these results. Disease-state data (Control/AD) is a domain-shift axis for conformal coverage, not clinical inference.

COMPETING INTERESTS

The author declares no competing interests.

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code. No AI system generated the scientific data, the evaluation outputs, or any reported number; every artifact was recomputed by the deterministic pipelines described in the Methods. All AI-assisted outputs were reviewed and verified by the author, who takes full responsibility for the content of the manuscript and the accompanying code. No AI tool is listed as an author.

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